Response Report for
A Near-Optimal Control Policy for Data-driven Assemble-to-order Systems

Submitted to *Operations Research (OR-2024-11-1514)*

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Dear Professor Ibrahim,

We submit our revised manuscript, "A Near-Optimal Control Policy for Data-driven Assemble-to-Order Systems," for possible publication in *Operations Research*. This paper is a revision of our previous submission (Manuscript ID: OR-2024-11-1514), originally submitted on November 26, 2024. We are grateful for the opportunity to revise the paper following the Major Revision decision received on June 9, 2025.

As summarized by the Associate Editor, the strengths of this paper are (i) tackling a practical and important problem of designing policies for assemble-to-order systems from data, and (ii) providing a strong sample complexity result. The concerns for this paper includes: (i) *Technical novelty*: the motivation for the convexity, the role of the asymmetric Lipschitz continuity (ALC) property, and the use/need for the decay approach are not clear or missing; (ii) *Algorithmic justification*: an experimental comparison to the Dynamic Programming algorithm is lacking; and (iii) *Exposition and clarity*: the writing and exposition need to be significantly improved.

We sincerely thank you and the entire review team for the constructive and insightful comments. We are deeply encouraged by the review team's assessment. Below, we highlight how the three major concerns are addressed.

(1) Substantiation of technical novelty. We clarify and strengthen the technical contributions by articulating the roles of (i) convexity, (ii) the ALC property, and (iii) damping (previously referred to as decay).

1. *Convexity.* We make the following three changes:

   - First, instead of using convexity in the whole reinforcement learning (RL) approach, we reconstruct the analysis in Section 4.2 (pp. 15–17) to restrict the use of convexity to an optimization subroutine of the reinforcement learning (RL) approach.
   - Second, we demonstrate that the standard Q-iteration may fail to converge in the absence of convexity in Appendix EC.3 (pp. ec32–34).
   - Third, we show in Section 4.6 (pp. 24–25) that enforcing convexity through an Input Convex Neural Network (ICNN) significantly accelerates convergence.

2. *ALC Property.* We make the following two changes:

   - First, we expand Section 4.5 (pp. 21–24) to formalize the Asymmetric Lipschitz Continuity (ALC) property by showing that, the ALC property is invariant under the Bellman operator and is therefore suitable for our analysis.
   - Second, we provide counterexamples to demonstrate that function classes defined by standard symmetric norms (e.g., L^1 or L^∞) are not invariant under the Bellman operator for ATO systems.

3. *Decay approach.* We have rewritten Section 4.1 (pp. 13–15) to provide a rigorous justification.

   - First, we clarify that the damped formulation ensures a bounded state space—a requirement that cannot be satisfied by simple truncation or restriction methods.
   - Second, we map the policies obtained from the damped problem to a feasible solution to the original, undamped ATO control problem.
   - Third, we guarantee that the damping modification neither changes the substantive decision problem nor compromises the correctness of the final results.

(2) Additional algorithmic benchmarking. Following referee 2's suggestion, we have implemented a Dynamic Programming (DP) benchmark using state aggregation (Section 5).

- First, the new experiments show that our CTD3 algorithm effectively mitigates the curse of dimensionality: it outperforms the DP benchmark in both *solution quality* and *computation time*.

- Second, the results confirm that CTD3 converges reliably on large-scale instances, whereas standard non-convex RL baselines consistently fail.

(3) Enhanced exposition and structural reorganization. We have substantially restructured the manuscript to improve clarity and strengthen the narrative flow. Two major changes are noteworthy:

- First, the contributions are positioned relative to the literature on ATO systems, data-driven optimization, and reinforcement learning.

- Second, we have reorganized Section 4 to present the theoretical analysis before the algorithmic implementation.

We thank you again for the opportunity to learn from the referees.

Best regards,

The authors

1 Response to the Associate Editor

We sincerely thank you for your insightful comments and constructive suggestions. We appreciate your encouraging comments. We have carefully revised the manuscript, paying particular attention to your suggestion on "*clarifying the technical contributions and reasoning of the paper, and sharpening the numerical experiments*." Our detailed point-by-point responses follow below.

1. **Comment:** *Both referees have significant concerns about the technical novelty of the paper. Since it appears that there are no serious issues with the correctness, I would like the authors to have the chance to address these concerns. Namely, R1 is concerned that the use of convexity is perhaps not well motivated.*

   Response: Thank you for the opportunity to revise the paper. We really appreciate this opportunity. In response to your summary that "*both referees have significant concerns about the technical novelty of the paper,*" we provide a high-level summary of our technical contribution:

   (a) *Theory development.* The contribution is three-fold:

   - First, our optimality gap analysis generalizes sample average approximation (SAA) from optimizing a single-valued, closed-form objective over a Euclidean space to optimizing function-valued objectives over infinite-dimensional spaces.
   - Second, the same result generalizes the empirical Markov decision process (EMDP) from finite state–action spaces to multidimensional and continuous spaces and handles the increased complexity.
   - Third, we utilize a novel asymmetric Lipschitz continuity (ALC) property to show that the optimal value function is Lipschitz continuous. Our discussion shows that the problem is challenging and that existing methods fail.

   (b) *Algorithmic development.* We propose a new benchmark algorithm with performance guarantees for the traditional assemble-to-order (ATO) control problem.

   - Theoretically, we prove that the proposed policy achieves near-optimal performance with limited historical demand data.

- Numerically, we compare the proposed policy with the existing heuristics and find notable performance improvement.

(c) *Convergence.* We propose a convex RL algorithmic design that achieves global convergence by exploiting the convex property of ATO control systems.

- Existing off-the-shelf RL algorithms cannot guarantee global convergence due to the potential non-convexity of the value function.
- To guarantee convergence, we equivalently simplify an important optimization subroutine in the traditional RL algorithm to a convex minimization program, where any local minimum is also a global minimum.

Next, we show each change:

(a) The role and motivation of convexity. In response to R1's comment that "*the use of convexity is perhaps not well motivated,*" we have discussed the implications of convexity as follows.

(i) *Limitation of the existing approaches.*

Our design and analysis focus on achieving a computationally tractable optimization of the Q function, which is a challenging constrained minimization problem over a continuous, multidimensional, non-orthotope feasible set defined by state-dependent constraints. In particular, existing off-the-shelf RL methods are not applicable as they require the action space to be a state-independent orthotope (see, for example, Fujimoto et al. 2018). As numerically illustrated in the revised paper, this optimization may be trapped at local minima or flat regions, causing Q-learning to converge to suboptimal values.

We provide more details in Section 4.2 of the revision (p. 16):

Our algorithmic design and analysis concentrate on a fast evaluation of $\mathcal{T}^* w_k$ at given state-action pairs...

(ii) *Convergence to suboptimality.*

We construct counterexamples in Appendix EC.3 to show that without convexity, the optimization subroutine can be trapped in local minima, causing Q-learning to converge to suboptimal value functions.

Our algorithmic design and analysis concentrate on a fast evaluation of $\mathcal{T}^* w_k$ at given state-action pairs, which requires an optimization subroutine to solve, for each $i \in [M]$, (...) This problem is challenging because the feasible set $\mathcal{A}_\alpha(\boldsymbol{s}')$ is not an orthotope and because (13) requires a global optimum. In particular, existing off-the-shelf RL algorithms, such as TD3 (Fujimoto et al. 2018), are not applicable without significant modification because they require the feasible set to be a state-independent orthotope. The convergence of such algorithms is also questionable because the optimization subroutine only guarantees local convergence and is easily trapped at local minima or flat regions, causing Q iteration to converge to a suboptimal value; an illustrative example is given in Appendix EC.3.

The specific counterexample is given in Appendix EC.3 (pp. ec35–36):

Example EC.2. *Consider a control problem with state space $\mathcal{S}_2 = \mathbb{R}_+$, state-dependent action space $\mathcal{A}_2(s) = [0, s+1]$ for $s \in \mathbb{R}_+$, and transition function $f_2(s,a) = s + a \ldots$ Consider concave function $Q_2(s,a) := s - a - 1 + \sqrt{(s+a+1)/2}$. It is straightforward to check that $a = s+1$ is a local minimizer of Q_2 for all $s \geq 0$, but not a global minimizer when $s \in [0,1]$, indicating that Q_2 is a suboptimal value. (...) Therefore, if a gradient descent algorithm that minimizes $Q_2(s,a)$ is stuck at the local minimum $a = s+1$, it causes the Bellman operator $\mathcal{T}_2^* Q_2$ to be falsely evaluated by Q_2, so that the Q iteration converges to a suboptimal value Q_2.*

(iii) *The benefits of convexity.* The revised Section 4.2 (pp. 15–17) now details the benefits of convexity:

- **Global convergence.** Convexity ensures that any local minimum of the optimization subroutine is also a global minimum.

  Conceptually, the convexity provides a significant simplification in the evalu-

ation of $\mathcal{T}^*$. For convex w, (13) simplifies to a convex program, i.e., minimizing a convex function over a convex polytope, where any local minimum is the global minimum. As a result, the evaluation $\mathcal{T}^* w(\boldsymbol{s}, \boldsymbol{a})$ is computationally tractable for any convex function w and $(\boldsymbol{s}, \boldsymbol{a}) \in \mathcal{SA}_\alpha$ using, for example, the interior point method (see Boyd and Vandenberghe 2004, Chapter 11).

– **Performance guarantee.** Convexity allows us to modify existing RL algorithms to achieve performance guarantees.

  Therefore, provided a sufficiently accurate convex *approximation to $\mathcal{T}^* w_k$, the computation of Q^*_α and, in turn, the near-optimal policy in Lemma 1(ii) becomes computationally tractable as well. In turn, Theorem 1 provides a conceptual justification that existing RL algorithms, with appropriate modifications, can be used to find near-optimal solutions to the DSAA control problem.*

– **Computational acceleration.** In Section 4.6 (pp. 24–25), we show that an Input Convex Neural Network (ICNN) can accelerate the algorithmic convergence.

  (First,) ICNN eliminates such minima by enforcing convexity. Therefore, we expect that CTD3 achieves accelerated convergence and improved stability. Second, the CTD3 algorithm sets to 0 the policy noise, which is an artificial Gaussian noise added to TD3 when evaluating actor value functions (see Equation (13) and (14) in (Fujimoto et al. 2018)) in order to stabilize the fit.... Setting it to 0 lets the actor exploit exact convex gradients, yielding faster and more stable learning, and simpler tuning.

  We also enriched the discussion in Section 5.3 (p. 32) that our proposed CTD3 algorithm outperforms a nonconvex RL benchmark.

  Our numerical experiments on multiple ATO systems clearly demonstrate the benefit of the convex design. Compared to the nonconvex RL benchmark, the proposed CTD3 algorithm converges faster in the small system and to a bet-

ter policy in the mid-size and large systems. It is also significant that the nonconvex RL benchmark fails to converge in the largest example, where the CTD3 algorithm converges over a wide range of learning rates. As the only modification in CTD3 from PTD3 is to approximate actor value functions by ICNN, our experiments thus provide a numerical justification for the substantial benefit of the convex algorithmic design.

(iv) *Representation power of ICNN.* We have generalized Lemma 2 in Section 4.6 (and Appendix EC.1) to prove that the input convex neural network (ICNN) uniformly approximates the convex Q-functions.

Lemma 2 (*Representation power of ICNN*) *On any compact domain, there exists an ICNN family* $\{Q_\phi : \phi \in \Phi\}$ *such that*

$$\inf_{\phi \in \Phi} \|Q - Q_\phi\|_\infty \leq \epsilon$$

holds for any bounded convex function Q*.*

. . . (This Lemma) implies that each Q function in Algorithm 1 can be uniformly approximated by a proper ICNN class to an arbitrary level of accuracy.

(b) Novelty of asymmetric Lipschitz continuity (ALC). In Section 4.5 (pp. 20–22), we have expanded the discussion to clarify why ALC is necessary and how it is utilized in our analysis.

(i) First, we have illustrated that function classes defined by standard symmetric norms (e.g., L^1 or L^∞) are not invariant under the Bellman operator. By contrast, the ALC property is invariant under the Bellman operator.

(ii) Second, we have expanded the exposition to clarify the intrinsic connection between ALC and the asymmetric production structure inherent to ATO systems.

(iii) Third, we have added a new bibliographic discussion highlighting that our use of ALC addresses a known challenging problem in analyzing empirical solutions for infinite-horizon MDPs on continuous state spaces.

Please refer to our response to Comment 3 below for further details on the ALC property.

2. **Comment:** *R2 doesn't see how transforming the problem from high-dimensional DP to high-dimensional NN is necessarily beneficial, without some more careful experiments and technical details.*

   Response: Thank you! We agree with the need to justify that "*transforming the problem from high-dimensional DP to high-dimensional NN is necessarily beneficial*." In the revised manuscript, we have added:

   (i) a heuristic justification for the NN approximation;

   (ii) a new comparative benchmark to DP-based state aggregation algorithm (Nadar et al. 2018); and

   (iii) extensive numerical experiments demonstrating the superior scalability and performance of the NN approach.

   The detailed revisions are as follows:

   (a) Applicability of NN. In Section 4.2 of the revision, we provide a heuristic explanation for the use of NNs.

   To specify, a Q learning algorithm approximates each w_k within a suitable function class $\mathcal{G}$. Typically, $\mathcal{G}$ a neural network (NN) class as it can uniformly approximate any continuous function on a compact set (Cybenko 1989). While obtaining such an approximation may be difficult in theory, especially when w_k is high-dimensional, the practical success of NN on control problems of similar dimensionality (see, for example, Gijsbrechts et al. 2022 and Dai and Gluzman 2022) supports its utility to our DSAA problem.

   (b) Difficulty of the DP benchmark. In Section 1.1, we clarify that the optimal policy lacks a simple structure (e.g., base-stock), making exact DP computationally prohibitive due to continuous, multidimensional state-action spaces.

However, the optimal policy is also known not to have a base-stock structure and is prohibitively hard to compute by dynamic programming, because the state and action spaces are continuous and multidimensional.

To establish a benchmark, we implemented an approximate DP benchmark in Section 5.1 based on the state aggregation method proposed by Nadar et al. (2018), which discretizes the state space for computational feasibility.

For the third benchmarking policy, we approximately solve the control problem using the state aggregation method proposed by Nadar et al. (2018). We discretize the state space using a finite grid. To form the grid, k vertices are sampled on each dimension, ... The dynamic programming (DP) algorithm is conducted by value iteration using an empirical demand distribution based on the M demand samples. Each iteration requires evaluating the state transition function over each aggregate state, action, and demand sample, counting for $k^{ps+n} \times m! \times M$ evaluations of state transitions.

(c) Mechanisms: targeted vs. exhaustive. In Section 5, we provide a heuristic justification for the benefit of the NN approach: it focuses its computational resources on relevant regions of the state space, whereas DP requires an exhaustive search.

Compared to a conventional DP solution that requires exhaustive traversal of the state space, RL algorithms focus on learning the subset of the most relevant states. It is already well recognized in reinforcement learning theory (see Chapter 9 of Sutton et al. 1998) that the targeted allocation of representational capacity in NN training not only reduces the computational burden compared to uniform DP sweeps, but also enhances accuracy in regions of the state space that most directly influence performance.

We further illustrate this by analyzing the complexity of the DP benchmark in Section 5.1, noting that even with coarse discretization, the computational cost becomes prohibitive.

With eight vertices in each dimension, there are 262,144 aggregated states. The reduced action space contains 128 actions. With 200 demand samples, each DP iteration *requires evaluating approximately* 10^{10} *state transitions, and the computational time easily exceeds that of the RL algorithms. Because our testing environment cannot accommodate further increases in the number of vertices per dimension due to computational capacity constraints, we conclude that the RL algorithms significantly outperform the state aggregation algorithm under reasonable computational power.*

Due to its computational complexity, the DP benchmark must rely on a small number of aggregate states, leading to substantial discretization error and, consequently, a significantly larger performance gap in cost minimization compared to RL algorithms.

It is worth noting that the computational complexity of the DP-based state aggregation method becomes astronomical even with only four vertices per dimension, with nearly 10^{16} *evaluations of state transition required for each value iteration, given* $M = 10$ *demand samples. Consequently, we only perform numerical experiments on the CTD3, PTD3, and NV-PRP methods.*

(d) Numerical evidence I: cost efficiency. In Section 5.2, we present results showing that the DP benchmark (referred to as the state aggregation algorithm in the revision) consistently underperforms the proposed RL algorithms, regardless of discretization resolution. This gap is attributed to the discretization error inherent in feasible DP implementations.

Despite a high computational complexity, the state aggregation algorithm performs poorly. With four vertices per dimension and $M = 10$*, the resulting policy yields a cost of 105.6, exceeding the optimal benchmark of* 81.0 *by over 30%. While increasing the resolution to six or eight vertices narrows this disparity, the optimality gap remains at 9.3% even with eight vertices and 200 demand samples. (Note that the proposed RL algorithm has a gap of 1.4%, according to Table 2 of the revision)*

(With correlated demands,) the performance of the DP-based state aggregation method is also consistent with its behavior in the i.i.d. setting, although it can outperform the NV-PRP policy by 7.9% under the finest discretization. Even with eight lattices, the DP policy remains less competitive than RL, achieving a cost of 67.9 for $M = 200$*, which*

is still significantly higher than those of CTD3 and PTD3, 58.1 and 61.1, respectively.

The corresponding numerical results are presented in Section 5.2 of the revision. For ease of reference, we also provide the results in Table 2 below.

	Algorithm	Demand Sample Size		
		$M = 10$	$M = 100$	$M = 200$
		C (s.d.)	C (s.d.)	C (s.d.)
I.I.D. Demand (Optimality: 81.0)	CTD3	86.9 (0.04)	82.9 (0.06)	82.2 (0.04)
	PTD3	87.4 (0.04)	84.6 (0.05)	83.0 (0.04)
	NV-PRP	86.1 (0.05)	84.7 (0.03)	82.9 (0.04)
	DP (4 Lattices)	105.6 (0.08)	105.5 (0.09)	105.3 (0.06)
	DP (6 Lattices)	100.8 (0.07)	100.9 (0.07)	97.4 (0.06)
	DP (8 Lattices)	99.7 (0.04)	88.9 (0.06)	88.5 (0.05)
Correlated Demand	CTD3	66.1 (0.03)	60.2 (0.05)	58.1 (0.02)
	PTD3	67.3 (0.03)	62.6 (0.04)	61.1 (0.05)
	NV-PRP	75.8 (0.06)	78.9 (0.07)	73.7 (0.07)
	DP (4 Lattices)	92.1 (0.04)	92.1 (0.06)	92.0 (0.04)
	DP (6 Lattices)	75.0 (0.07)	75.0 (0.03)	74.9 (0.05)
	DP (8 Lattices)	70.4 (0.05)	70.4 (0.04)	67.9 (0.03)

Table 2: The performances of CTD3 policy, PTD3 policy, NV-PRP policy, and DP policies on the N-system with different demand sample sizes and demand joint distributions.

(e) Numerical evidence II: computational complexity. We demonstrate that the RL algorithms are not only more accurate but also more scalable. As shown in Table EC.3, the training time for the DP benchmark escalates rapidly with resolution, eventually exceeding the RL training time while still yielding inferior policies.

We observe that the DP benchmark is computationally efficient at coarse discretization levels; however, its training time escalates rapidly as the resolution (number of lattice points) increases, eventually exceeding that of the RL algorithms. Furthermore, the CTD3 algorithm converges significantly faster than PTD3, empirically validating the role of convexity in accelerating convergence. Note that we do not report training time for the NV-PRP heuristic, as it does not require training.

3. **Comment:** *R2 is also unsure about the ALC property. I was also surprised to see it, since it is a standard issue in inventory management.*

	Algorithm	Demand Sample Size		
		$M = 10$	$M = 100$	$M = 200$
Training Time (s)	CTD3	2482	4035	4420
	PTD3	4980	6311	9856
	DP (4 Lattices)	53	385	738
	DP (6 Lattices)	1434	10937	19137
	DP (8 Lattices)	14012	107580	221687

Table EC.3: Training time comparison for N-system. Inference time is averaged over 1,000 steps.

Response: We appreciate your insight regarding the ALC property. Our goal is to prove Lipschitz continuity of the value function for an infinite-horizon problem, which is not possible using existing theory.

(a) The difficulty. In the revised manuscript, we have added additional justification for why Lipschitz continuity is challenging to prove.

(i) *A bibliographic remark.* We first provide a bibliographic remark on this question.

...we remark that the need for Lipschitz continuity is known in analyzing empirical solutions for infinite horizon MDP on continuous state spaces, see, for example, Li et al. (2021) and Gupta et al. (2024). A sufficient condition for such Lipschitz continuity is given by Hinderer (2005), which essentially specifies that an L-Lipschitz family with respect to a proper norm is an invariant subset of the Bellman operator.

(ii) *A counterexample.* To demonstrate that these standard arguments do not apply, we have added specific counterexamples in Section 4.5. These examples show that function classes defined by standard symmetric norms (like L^1 or L^∞) are not invariant under the Bellman operator for ATO systems.

Example 1 *Consider an ATO system in which a single product is assembled from two components, each with a unit lead time. Assembly takes one unit of each component. There is no arriving demand. Holding and backlog costs are 1. The state and action are* $\boldsymbol{s} := ((x_1, x_2), \tilde{u}, (R_{1,1}, R_{2,1}))$ *and* $\boldsymbol{a} := (y, (z_1, z_2))$. *We consider a discount factor* $\gamma = 0.9$ *and a damping factor* $\alpha = 0.1$. *Take demand* *support* $\bar{d} = 1$ *so that* $(\boldsymbol{s}, \boldsymbol{a}) \in \mathcal{SA}_\alpha$ *whenever* $\|\boldsymbol{s}\|_\infty \leq 10$, $z_1, z_2 \in [0, 10]$, *and* $y \in [0, \min\{x_1, x_2, \tilde{u}\}]$.

In Example 1, Theorem 3 fails under commonly used norms and Lipschitz continuity.... Define function $w(\boldsymbol{s}, \boldsymbol{a}) := L(x_1 + x_2 + \tilde{u} + y)$, *which is* $(L, \|\cdot\|_1)$*-Lipschitz and* $(4L, \|\cdot\|_\infty)$*-Lipschitz. ... Therefore, there exists no* L *such that* $\mathcal{T}^* w$ *and* w *have the same Lipschitz constant as* w*. In particular, we cannot infer, using the existing approach, that the function* Q^*_α*, which is the unique fixed point of* $\mathcal{T}^*$*, must belong to a Lipschitz class.*

(b) Necessity of asymmetry. In the revision, we have made additional efforts to show that the asymmetry is necessary.

(i) *An additional counterexample.* The revision has added a counterexample in Section 4.5 to indicate that a symmetric bound is not sufficient.

To further illustrate the role of asymmetry, consider seminorm

$$\rho_1'(\boldsymbol{s}, \boldsymbol{a}) := \|\boldsymbol{x} - A\boldsymbol{y}\|_1 + \|\tilde{\boldsymbol{u}} - \boldsymbol{y}\|_1 + \sum_{i=1}^{n} \|R_i\|_1 = |x_1 - y| + |x_2 - y| + |\tilde{u} - y| + |R_{1,1}| + |R_{2,1}|,$$

which is the symmetric version of ρ_1*. However, the Lipschitz constant with respect to* ρ_1' *again exhibits no contraction under* $\mathcal{T}^*$*. To show this, take* $w'(\boldsymbol{s}, \boldsymbol{a}) = L(x_2 - x_1 + \tilde{u} - y)\ldots$

(ii) *Intuitions.* We provide clearer justifications in Section 4.5 that a valid invariant subset must have asymmetric bounds on the slopes along different directions.

There is an intrinsic explanation that the ALC property in Theorem 3 captures the asymmetry between overage and underage in assembly systems. Specifically, in some states, a decrease in the inventory of any component can cause insufficient production and, in turn, an increase in the inventory of many other components in the next period. Therefore, among all functions in an invariant subset of the Bellman operator, the maximum slope evaluated at such states must be higher in the direction of decreased inventory compared to the direction of increased inventory, *necessitating an asymmetry.*

(c) Sufficiency. The revision has also provided more details on how the ALC property is used to establish the desired Lipschitz continuity, thereby justifying its sufficiency.

It is therefore significant from Theorem 3 that, with such asymmetry captured, a proper ALC family becomes an invariant of the Bellman's operator. Using the iteration in Lemma 1, the function Q^*_α is ALC and, in turn, Lipschitz continuous. Using Lemma 1(iii), we can then prove the Lipschitz continuity of $V^*_\alpha(\,\cdot\,;\widehat{F})$. Our novel finding here is that the Lipschitz continuity, which is a symmetric notion, needs to be proved by accounting for the asymmetric behavior of the ATO system.

4. **Comment:** *R1 and R2 also have concerns about the use / need of the decay approach.*

   Response: We appreciate the reviewers' scrutiny regarding the decay (now renamed "damping") approach. The review team has raised questions about (a) whether the use of the damping (we previously called decay) has shifted the original problem to a different one, and (b) whether the damping approach is necessary. We have addressed both points, as elaborated below.

   (a) The use of damping. The main changes are summarized below.

   (i) In Section 4.1 (pp. 13) of the revised manuscript, we have added the following sentence.

   We show in Section 4.4 below that, by taking $\alpha = O(M^{-1/2})$ with M being the demand sample size, the error caused by damping and ceiling is asymptotically negligible as $M \to \infty$.

   At the end of Section 4.1 (pp. 14), we have added the following additional clarifications.

   Our goal is to find and transform the minimizer (of the damped control problem) to a data-driven policy of the original ATO control problem.

(ii) In Section 4.3 (pp. 17) of the revised manuscript, we have added further explanation of the algorithmic design and emphasized that the resulting policy is "translated" back to the original, undamped control problem.

...step 5 of Algorithm 1 translates the map ξ_K into a data-driven policy of the undamped ATO control problem.

(iii) Finally, we have added further explanations in Section 4.4 (pp. 19) of the revised manuscript regarding the main technical result, including a clarification that the optimality gap is evaluated for the original, undamped control problem.

Here, the optimality gap is measured with the undamped and unknown ATO dynamics, not the DSAA dynamics.

(b) The need for damping. In the revised manuscript (Section 4.1), we have explained why the damping factor is needed.

(i) *Preservation of structural properties.* To apply the necessary concentration inequalities in our main proof, we require a bounded state space. However, we need to ensure that the system retains **linearity** in transition dynamics and **convexity** in the state-action space. As detailed in the expanded Proposition 1 in Section 4.1 (pp. 13–14), the damping approach satisfies all these three requirements:

Proposition 1 (Boundedness) *The following results hold....*

Proposition 1 suggests that the damped transition function f_α preserves the linearity of f and the damped state-action space $\mathcal{SA}_\alpha$ preserves the convexity of $\mathcal{SA}$. These two properties are crucial in subsequent results.

(ii) *Incompatibility of standard alternatives.* We actively explored standard alternatives to the damping-factor approach. Specifically, we find that the following approaches do not work.

- *Truncation.* Forcing a hard bound on the state space (e.g., $\min\{f(\cdot), K\}$) ensures boundedness but destroys the **linearity** of the transition function f.

– *Action restriction.* Restricting actions to "reasonable" subsets (as in Reiman and Wang (2015)) can bound the trajectory, but it results in a **non-convex** state-action space.

For example, truncating the state transition by $\boldsymbol{s}(t+1) = \min\{f(\boldsymbol{s}(t), \boldsymbol{a}(t), \boldsymbol{d}(t)), K\}$ *for some* $K > 0$ *yields a bounded state space but destroys the linearity of* f*. Restricting actions to "reasonable" ones, as in Reiman and Wang (2015), can bound the state trajectory under the optimal policy but may lead to a non-convex state-action space.*

For the aforementioned reasons, we retain the damping factor. More details are provided in Section 4.1 of the revised manuscript.

5. **Comment:** *There is general concern that the technical results are more standard than innovative. I believe that is always great to leverage existing tools as much as possible, and I'd like the authors to clarify the difficult/technical steps needed beyond the previous known results. Perhaps providing counterexamples showing existing tools are not enough at key proof steps might help.*

   Response: Thank you, and we have followed your suggestions as follows:

   (a) Theoretical difficulty.

   (Our result analyzes) how the objective, which is function-valued, without a closed-form, and minimized over an infinite-dimensional space, is sensitive to a multidimensional unknown distribution. Indeed, the literature does not provide guidance on how to assess such sensitivity.

   We contribute to the literature on sample average approximation (SAA) and empirical Markov decision process (EMDP).

   We make additional methodological contributions to data-driven optimization. Our sample-efficiency analysis of ATO systems extends SAA from optimizing a single-valued, closed-form objective over a Euclidean space (see, for example, Levi *et al. 2007 and Cheung and Simchi-Levi 2019) to optimizing a function-valued objective without a closed-form over an infinite-dimensional space. Our results also establish, for the first time, a nonasymptotic gap for infinite-horizon empirical Markov decision process (EMDP, see Kalathil et al. 2021) with multidimensional continuous state and action spaces, whereas existing results are limited to finite (in cardinality) spaces.*

We propose a new, asymmetric form of Lipschitz continuity to establish Lipschitz continuity of the optimal value function.

A central element of our analysis is the ALC property, which provides a new tool for regularizing value functions in existing results (see, for example, Hinderer 2005) and has the potential to be extended to other control problems in inventory, production, and queueing management.

(b) *Algorithmic difficulty.* We list our changes in the introduction section as below.

existing off-the-shelf RL methods are not applicable as they require the action space to be a state-independent orthotope (see, for example, Fujimoto et al. 2018). As we show in this paper, the optimization subroutine may also be trapped at local minima or flat regions, causing Q-learning to converge to a suboptimal value.

We have emphasized that the algorithmic design supports *global convergence* to a near-optimal policy, owing to the problem's convexity.

our algorithmic design provides new tools for the application of RL to continuous control problems with state-dependent action spaces. (...) Exploiting the convex property, the use of IPM supports a global convergence to a near-optimal solution.

We have emphasized that the proposed algorithm outperforms existing policies and an adapted RL benchmark.

The convex property justifies the benefits of adopting convex approximation to

> *convex Q functions, which is applicable through existing shape-constrained network architectures. (...) Our numerical results demonstrate the relative effectiveness of the specialized RL algorithm compared with existing heuristic policies, dynamic programming algorithms, and a minimally modified RL benchmark.*

Following your recommendation to "*providing counterexamples showing existing tools are not enough at key proof steps might help,*" we have revised Sections 4.2 and 4.5 to give five counterexamples that serve different but all crucial purposes:

- The first three are associated with Example 1, which mathematically demonstrates that standard Lipschitz-based tools (which work in other settings) fail to remain invariant under the Bellman operator in our problem. This failure necessitates the development of the ALC property.
- Examples 2 and 3 demonstrate the necessity of convexity, where the optimization subroutine may otherwise be trapped at local minima and cause Q iteration to stop converging.

For the detailed counterexamples, please refer to Sections 4.2 (pp. 14–16) and 4.5 (pp. 20–22) of the revised manuscript and our response to Comments #1 and #3.

6. **Comment:** *Both reviewers give suggestions on writing improvements.*

   Response: Thank you. We have carefully incorporated the referees' suggestions to improve the clarity and quality of the writing throughout the manuscript.

2 Response to Referee 1

Thank you for your insightful comments and encouragement. Guided by your feedback, we have revised the manuscript as follows:

1. **Comment:** *The main point of the paper is to formulate the problem as a reinforcement learning problem with some convex structure, and use a neural network to approximate the Q function, a fairly standard approach in deep Q-Learning.*

   Response: Thank you for leading us to delve deeper into the core contributions of the paper. By carefully examining the contribution of the paper, we respectively disagree with you that the approach is "*a fairly standard approach in deep Q-Learning*" for the following two reasons.

   (i) Algorithmic difference. Off-the-shelf reinforcement learning (RL) algorithms cannot be directly applied to our assemble-to-order (ATO) control problem because its has a continuous and state-dependent action space that is not an orthotope. The global convergence of such algorithms are not guaranteed because they lack tools for globally minimizing over the action space.

   (ii) Theoretical difference. Unlike the standard RL algorithms, where the transition is a black box, the transition of our ATO problem is known up to a finite-dimensional demand distribution. Due to this difference, our main technical result, the *sensitivity* over unknown demand distribution, requires a new proof and is important to the literature on ATO systems.

   We would also like to clarify that our main points of the paper extend beyond "*formulating the problem as an reinforcement learning problem with some convex structure.*" Our goal is to answer the following two research questions, which we have revised as follows (see pp. 2–4 of the revised manuscript).

   (i) *The first research question is: Assuming infinite computational power, what is the performance gap of the best control policy given a demand sample of size M, and what is its growth rate with respect to the state dimension p?*

(ii) *The statistical tractability of the ATO control problem immediately motivates the second research question: Is a near-optimal data-driven policy of the ATO control problem computationally solvable? What is the computational complexity?*

Meanwhile, we have made the following related revisions:

(a) Analytical framework. We have revised the analytical framework as follows. See also pp. 2-4 of the revised manuscript.

(i) Regarding the first research question, the revision is as follows.

(We analyze) how the objective, which is function-valued, without a closed-form, and minimized over an infinite-dimensional space, is sensitive to a multidimensional unknown distribution. Indeed, the literature does not provide guidance on how to assess such sensitivity.

(ii) Regarding the second research question, the revision is as follows.

Our design and analysis focus on achieving a computationally tractable optimization of the Q function, which is a challenging constrained minimization problem over a continuous, multidimensional, non-orthotope feasible set defined by state-dependent constraints. In particular, existing off-the-shelf RL methods are not applicable as they require the action space to be a state-independent orthotope (see, for example, Fujimoto et al. 2018).

(b) Our contributions. We have revised the paper's contributions as follows.

(i) *Insights into ATO systems.*

Our results show that, with a finite historical demand sample, a data-driven policy attains a near-optimal performance with a gap that compares only logarithmically larger than that between the sample mean and the unknown mean demand.

(ii) *Benchmark development.*

The algorithm establishes a new benchmark for the ATO control problem and is practically implementable using demand samples rather than exact distributions.

A convex property of the ATO transition dynamics justifies the computational tractability. Numerical results demonstrate its effectiveness over existing heuristic policies and dynamic programming algorithms, which also suggest that a widely adopted no-holdback rule, which is optimal for single-product ATO systems, can incur double-digit optimality losses in multi-product ATO systems.

(iii) *Technical novelty.*

Our sample-efficiency analysis of ATO systems extends sample average approximation (SAA) from optimizing a single-valued, closed-form objective over a Euclidean space (see, for example, Levi et al. 2007 and Cheung and Simchi-Levi 2019) to optimizing a function-valued objective without a closed form over an infinite-dimensional space. Our results also establish, for the first time, a nonasymptotic gap for infinite-horizon empirical Markov decision process (EMDP, see Kalathil et al. 2021) with multidimensional continuous state and action spaces, whereas existing results are limited to finite (in cardinality) spaces. A central element of our analysis is the asymmetric Lipschitz continuous (ALC) property, which provides a new tool for regularizing value functions in existing results (see, for example, Hinderer 2005) and has the potential to be extended to other control problems in inventory, production, and queueing management.

(iv) *RL applicability.*

Finally, our algorithmic design provides new tools for the application of RL to continuous control problems with state-dependent action spaces. The convex property justifies the benefits of adopting convex approximation to convex Q functions, which is applicable through existing shape-constrained network architectures. Exploiting the convex property, the use of interior point method (IPM) supports a global convergence to a near-optimal solution. Our numerical results demonstrate the relative effectiveness of the specialized RL algorithm compared with existing heuristic policies, dynamic programming algorithms, and a minimally modified RL benchmark.

(c) Nonstandard solution approach.

(i) *Algorithmic difficulty.* In Section 4.2, we have emphasized that no existing off-the-shelf RL algorithms can be directly applied to the ATO control problem.

(An off-the-shelf algorithm requires a fast) optimization subroutine to solve, for each $i \in [M]$,

$$\min_{\boldsymbol{a}' \in \mathcal{A}_\alpha(\boldsymbol{s}')} w(\boldsymbol{s}', \boldsymbol{a}'), \quad s' = f_\alpha(\boldsymbol{s}, \boldsymbol{a}, \boldsymbol{d}_i).$$

This problem is challenging because the feasible set $\mathcal{A}_\alpha(\boldsymbol{s}')$ is not an orthotope and because (13) requires a global optimum. In particular, existing off-the-shelf RL algorithms, such as TD3 (Fujimoto et al. 2018), are not applicable without significant modification because they require the feasible set to be a state-independent orthotope. The convergence of such algorithms is also questionable because the optimization subroutine only guarantees local convergence and is easily trapped at local minima or flat regions, causing Q iteration to converge to a suboptimal value; an illustrative example is given in Appendix EC.3.

(ii) *Theoretical novelty.* In Section 4.5, we explain that our main technical result cannot be proved by existing methods.

Our analysis, through existing concentration inequalities, reduces the sensitivity analysis for the ATO control problem to proving Lipschitz continuity of the optimal value function. Unfortunately, it is known that the Lipschitz continuity is difficult to prove for infinite-horizon control problems. As a general theory, Hinderer (2005) provides sufficient conditions, supporting analyses like Gupta et al. (2024) and Li et al. (2021). However, the conditions are restrictive and inapplicable to ATO systems. As a novel result, we show that the Bellman operator of the ATO control problem admits an unusual ALC class as a nontrivial invariant...

2. **Comment:** *The authors argue that the convexity of the Q function makes it amenable to approximation by the class of input convex neural networks. However, the main justification the authors provide for this statement appears to be that such neural networks are capable of approximating any convex function. This argument seems unjustified,*

especially since the authors assume an oracle which approximates the Q-function uniformly over its entire domain.

Response: Thank you for your valuable feedback. We acknowledge that our initial justification was inadequate. In the revised manuscript, we have discussed (i) the specific choice of input convex neural networks (ICNN) and (ii) the benefits of the convexity property as follows.

(a) The representation power of ICNN. In the revised manuscript, we have proved a technical lemma on the representation power of an ICNN.

Lemma 2 (Representation power of ICNN) *On any compact domain, there exists an ICNN family $\{Q_\phi : \phi \in \Phi\}$ such that*

$$\inf_{\phi \in \Phi} \|Q - Q_\phi\|_\infty \le \epsilon$$

holds for any bounded convex function Q.

This result extends the prior work of Chen et al. (2019) (which established approximation accuracy for Lipschitz convex functions) by leveraging the separating hyperplane theorem.

(b) Compact domain of the Q functions. In the revised manuscript, we show that the state-action space $\mathcal{SA}_\alpha$ is compact in Proposition 1. See Section 4.2 (pp. 15) for the details:

Notice that w_k and Q^*_α have a common domain $\mathcal{SA}_\alpha$, which, according to Proposition 1, is compact.

Furthermore, we have discussed the approximation of a Q-function using an ICNN. This addition (in Section 4.6) links our theoretical results to the practical implementation, justifying the use of the ICNN in Algorithm 1.

Given a damping factor $\alpha > 0$, the DSAA control problem has a compact domain, implying that each Q function in Algorithm 1 can be uniformly approximated by a *proper ICNN class to an arbitrary level of accuracy. Therefore, ICNN exemplifies a valid choice of operator $\mathcal{N}$ in Algorithm 1, where for any convex function g on $\mathcal{SA}_\alpha$, $\mathcal{N}(g)$ can be taken as a fit of g using the ICNN family.*

(c) Additional clarification. To reduce ambiguity, we have removed some distracting discussions on uniform approximation and the noncompactness of the ATO system.

Finally, we have clarified in Section 4.6 that the convexity property serves a deeper theoretical purpose than "*makes it amenable to approximation by input convex neural networks.*" As elaborated in our response, the property and the use of ICNN together support our goal: the global convergence in the evaluation of the Bellman operator. The use of ICNN is not the goal here. We apologize for the confusion.

3. **Comment:** *I also do not see a good reason for convexity to help here: the covering-number size for the class of Lipschitz convex functions on a domain typically exhibits exponential asymptotic growth in the dimension. This is the same as the larger class of all Lipschitz functions (so the convexity property seems irrelevant for the approach in this paper).)*

   Response: We sincerely thank you for this careful observation. We recognize the need to provide a stronger justification for "*convexity to help here.*" Below, we first respond to the discussion regarding the covering number, and then provide a high-level summary of the benefits of the convexity property.

   (a) The covering number. We fully agree that the covering number for "*the class of Lipschitz convex functions*" grows exponentially with both the accuracy level and the dimension, which is a common issue in RL algorithms. In the revision, we now emphasize that the convexity property is **not** intended to overcome this curse of dimensionality by substantially reducing the search space of Q-functions.

   However, we respectfully note that there is a significant theoretical distinction between the complexity of the convex class and that of the general Lipschitz class:

   (i) It is shown in Bronshtein (1976), Theorem 6 that the logarithm of the ε-covering number for L-Lipschitz convex class on $[a, b]^p$, with respect to the uniform norm, is $O(\varepsilon^{-p/2})$. The exponential growth indicates a curse of dimensionality, reflecting the computational challenges of the ATO control problem.

(ii) The same ε-covering number, again after taking logarithm, is $O(\varepsilon^{-p})$ for "*the larger Lipschitz class of all Lipschitz functions*", see EXAMPLE 5.10 in Wainwright (2019).

(iii) The *difference in the exponential growth rate* is significant because, conceptually, the computation power required for approximating a Lipschitz convex function with dimension p is similar to that of a Lipschitz function with dimension $p/2$. Our numerical experiments indicate that this reduction is meaningful.

(b) Reasons that the convexity helps. In the revision, we have updated the analytical framework of the proposed Q-learning-based algorithm. Within the new framework, the revised manuscript now summarizes the benefits of the convex property. The detailed changes are as follows.

(i) *Where the convexity helps.* We have updated the analytical framework to better explain where the convexity can help.

Our algorithmic design and analysis concentrate on a fast evaluation of $\mathcal{T}^* w_k$ at given state-action pairs, which requires an optimization subroutine to solve, for each $i \in [M]$,

$$\min_{\boldsymbol{a}' \in \mathcal{A}_\alpha(\boldsymbol{s}')} w(\boldsymbol{s}', \boldsymbol{a}'), \quad s' = f_\alpha(\boldsymbol{s}, \boldsymbol{a}, \boldsymbol{d}_i).$$

This problem is challenging because the feasible set $\mathcal{A}_\alpha(\boldsymbol{s}')$ is not an orthotope and because (13) requires a global optimum. In particular, existing off-the-shelf RL algorithms, such as TD3 (Fujimoto et al. 2018), are not applicable without significant modification because they require the feasible set to be a state-independent orthotope.

(ii) *Why the convexity helps.* We explain why convexity simplifies the algorithm.

Conceptually, the convexity provides a significant simplification in the evaluation of $\mathcal{T}^*$. For convex w, (13) simplifies to a convex program, i.e., minimizing a *convex function over a convex polytope, where any local minimum is the global minimum. As a result, the evaluation $\mathcal{T}^* w(\boldsymbol{s}, \boldsymbol{a})$ is computationally tractable for any convex function w and $(\boldsymbol{s}, \boldsymbol{a}) \in \mathcal{SA}_\alpha$ using, for example, the interior point method (see Boyd and Vandenberghe 2004, Chapter 11).*

(iii) *The necessity of convexity.* The revision has substantially enriched the discussion in Section 4.2, showing that the convexity property is necessary for global convergence of Q-iteration algorithms that use an optimization subroutine via gradient descent.

The convergence of such algorithms is also questionable because the optimization subroutine only guarantees local convergence and is easily trapped at local minima or flat regions, causing Q iteration to converge to a suboptimal value; an illustrative example is given in Appendix EC.3.

To further justify the necessity, we provide two counterexamples; see Section 4.2 and Appendix EC.3 (pp. ec33–35). In both examples, the optimization of the Q-function becomes trapped at local minima, preventing the Q-iteration from converging to the optimal value function. In the first example, the Bellman operator lacks the convexity-preservation property. In the second example, the Q-function is not enforced to be convex.

Example EC.1. *Consider a control problem with state space $\mathcal{S}_1 = [0, 2]$, action space $\mathcal{A}_1 = [0, 1]$, and transition function $f_1(s, a) = s/2 + a$. Consider the control problem with the objective function*

$$\sum_{t=0}^{\infty} \gamma_1^t c_1(s(t), a(t)),$$

subject to $s(t+1) = f_1(s(t), a(t))$, which is minimized over $\{a(t) : t \geq 0\}$. Here $c_1(s, a) := 3s/4 + 5a/2 - 2a^2 - 1/2$ is not a convex function and $\gamma = 1/2$.

In the example, the Bellman operator $\mathcal{T}_1^*$ is given by

$$\mathcal{T}_1^* w(s,a) := c_1(s,a) + \gamma_1 \min_{a' \in [0,1]} w(f_1(s,a), a').$$

Consider concave function $Q_1(s,a) := s + 3a - 2a^2$. It is clear that Q_1 is a suboptimal value and $\mathcal{T}^* Q_1 < Q_1$. However, $a = 1$ is a local (but not global) minimum of $Q_1(s,a)$ on $a \in [0,1]$ and it is easy to check that

$$Q_1(s,a) = c_1(s,a) + \gamma_1 Q_1(f_1(s,a), 1), \quad \textit{for all } (s,a) \in \mathcal{S}_1 \times \mathcal{A}_1. \tag{1}$$

Therefore, if a gradient descent algorithm that minimizes $Q_1(s,a)$ is stuck at the local minimum $a = 1$, it causes the Bellman operator $\mathcal{T}_1^* Q_1$ to be evaluated by Q_1, so that the Q iteration converges to a suboptimal value Q_1.

Example EC.2. *Consider a control problem with state space $\mathcal{S}_2 = \mathbb{R}_+$, state-dependent action space $\mathcal{A}_2(s) = [0, s+1]$ for $s \in \mathbb{R}_+$, and transition function $f_2(s,a) = s + a$. The objective function is*

$$\sum_{t=0}^{\infty} \gamma_1^t c_2(s(t), a(t)),$$

subject to $s(t+1) = f(s(t), a(t))$, and is minimized over $\{a(t) : t \geq 0\}$ with $a(t) \in [0, s(t)+1]$. Here $c_2(s,a) := s - a - 1$ is a convex function and $\gamma = \sqrt{2}/2$. However, consider concave function $Q_2(s,a) := s - a - 1 + \sqrt{(s+a+1)/2}$. It is straightforward to check that $a = s+1$ is a local minimizer of Q_2 for all $s \geq 0$, but not a global minimizer when $s \in [0,1]$, indicating that Q_2 is a suboptimal value. However, we can again check that

$$Q_2(s,a) = c_2(s,a) + \gamma_2 Q_2(f_2(s,a), s+1), \quad \textit{for all } s \in \mathbb{R}_+ \quad \textit{and} \quad a \in [0, s+1]. \tag{2}$$

Therefore, if a gradient descent algorithm that minimizes $Q_2(s,a)$ is stuck at the local minimum $a = s+1$, it causes the Bellman operator $\mathcal{T}_2^* Q_2$ to be falsely evaluated by Q_2, so that the Q iteration converges to a suboptimal value Q_2.

(iv) *Convergence acceleration.* Finally, we have discussed how convexity contributes to the development of a better algorithmic design.

The requirement that $\mathcal{T}^* w_k$ be approximated by a convex function at every iteration can be practically achieved by existing shape-constrained NN architectures, see Section 4.6 below. As further discussed there and numerically demonstrated in Section 5.3, enforcing convexity yields substantial computational advantages compared to using a nonconvex NN.

In Section 4.6 (pp. 23–24) of the revision, we clarify how convexity, implemented through ICNN, leads to significant acceleration of convergence.

There is a clear benefit of employing a convex fit to the convex action-value functions via ICNN. To specify, a standard fully connected network can learn a wavy, nonconvex surface with false local minima and flat plateaus, which can make the algorithm settle on suboptimal actions and render optimization brittle and hard to converge. In Appendix EC.3, we provide a counterexample that false local minima due to a nonconvex fit can cause the Q iteration, i.e., Lemma 1(ii), to converge to a suboptimal value. In contrast, ICNN eliminates such minima by enforcing convexity. Therefore, we expect that CTD3 achieves accelerated convergence and improved stability. We also expect that those advantages are amplified in high-dimensional problems, where nonconvex neural fits are more likely to have local minima and flat plateaus.

Additionally, the convexity property allows us to set the policy-noise parameter to zero in the policy-gradient algorithm, which we adopt in our practical algorithmic design to achieve faster and more stable convergence.

(The policy noise) is an artificial Gaussian noise added to TD3 when evaluating actor value functions (see Equation (13) and (14) in (Fujimoto et al. 2018)) in order to stabilize the fit. With a convex Q fit via an ICNN, this noise is unnecessary and adds an extra source of bias and an extra hyperparameter. Setting it to 0 lets the actor exploit exact convex gradients, yielding faster and more stable

learning, and simpler tuning.

4. **Comment:** *Thus, I do not see any principled basis for the assumptions and framework in this paper. This is my biggest concern for the paper.*

   Response: Thank you for raising this fundamental concern. We recognize that the previous manuscript did not sufficiently articulate the theoretical and practical motivations driving our framework. In the revision, we have substantially rewritten the Introduction and main body to explicitly establish the principled basis for our approach. This basis rests on two core pillars:

   (a) Theoretical basis: finite-sample uncertainty. A central objective of this work is to analyze the ATO control problem under the realistic constraint of having only a finite sample of historical demand. The framework is designed to bridge the gap between standard inventory theory and the literature on Empirical MDPs. The value of our main result (Theorem 2) lies in quantifying how finite-sample uncertainty propagates through the system—a contribution that provides specific statistical insights not available in standard infinite-horizon analyses. See also our response to Comment 1.

   (b) Algorithmic basis: Bellman evaluation. Our assumption and analytical framework on the algorithmic design targets the computational challenge of Bellman evaluation in implementing Q learning based RL algorithms, which is specific to the ATO system. Our focus on this specific challenge justifies the assumption of access to "*an oracle that uniformly approximates the Q-function*" to simplify the theoretical challenge common to RL applications. Instead, our focus on Bellman evaluation has motivated a strong utility of the convex property, in terms of providing a conceptual justification, supporting implementable algorithmic design, and accelerating convergence substantially. See our response to Comment 3 above.

5. **Comment:** *The above idea appears to be the only new idea in the paper. Without the above idea, the Q-function-based RL approach used in the paper is standard. It is unfortunate that the only new idea does not seem to be justified.*

   Response: Thank you for your valuable suggestions. We have rewritten Section 1 (pp. 4–5) to highlight our contributions. As summarized below (see also our response to Comment 1):

(a) *Research question.* We study a new and practically realistic research question: how the ATO control problem is affected by demand learning.

(b) *RL benchmark for ATO.* It is of practical importance to establish a new RL-based benchmark for the ATO control problem.

(c) *Methodological contributions.* Our analysis includes nontrivial methodological advances in generalizing SAA techniques to data-driven control problems and in extending EMDP methods to multidimensional and continuous state–action spaces.

(d) *Proof technique.* The proof introduces the nonstandard ALC property to establish Lipschitz continuity of the optimal value function. The ALC property captures the asymmetric structure between overage and underage costs that is common in inventory and production systems.

For concision, we do not repeat the revisions made; all changes are addressed in our responses to the earlier comments.

6. **Comment:** *Technically speaking, the paper changes the problem that they initially introduced by adding a decay factor $(1 - \alpha)$ to ensure finiteness of the infinite horizon value function.*

Response: We apologize for this confusion. The decay factor is merely a solution approach. Our goal is always the original problem.

(a) In Section 4.1 (pp. 13) of the revised manuscript, we have clarified at the point where the damping factor is first introduced.

We show in Section 4.4 below that, by taking $\alpha = O(M^{-1/2})$ with M being the demand sample size, the error caused by damping and ceiling is asymptotically negligible as $M \to \infty$.

(b) At the end of Section 4.1 (pp. 14), we have emphasized that

Our goal is to find and transform the minimizer (of the damped control problem) to a data-driven policy of the original ATO control problem.

(c) In Section 4.3 (pp. 17) of the revised manuscript, we have emphasized that the resulting policy is "translated" back to address the original, undamped control problem.

...step 5 of Algorithm 1 translates the map ξ_K into a data-driven policy of the undamped ATO control problem.

(d) Finally, we have emphasized in Section 4.4 (pp. 19) that the optimality gap is evaluated for the original, undamped control problem.

Here, the optimality gap is measured with the undamped and unknown ATO dynamics, not the DSAA dynamics.

7. **Comment:** *The main theoretical result is Theorem 2, which gives a regret bound for their Algorithm 1 under suitable assumptions.*

Response: Theorem 2 has practical impact and implications for ATO systems. Thank you for encouraging us to make a highlight. In response, we have revised Section 4.4 (pp. 18–19) to present these implications more explicitly.

Theorem 2 demonstrates the statistical tractability of the data-driven ATO control problem. (...) Taking $\varepsilon_m = O(M^{-1/2})$, Theorem 2 shows that the optimality gap of the best data-driven policy and the unknown optimal policy is also $O(M^{-1/2}\log M)$. (...) The gap is close to $O(M^{-1/2})$—the difference between the sample mean and population mean. Therefore, although the optimal value function $V^*(\,\cdot\,; F)$ depends on the distribution F more intricately than the mean $\mathbb{E}[\boldsymbol{d}]$, the sample size required to accurately approximate the former is only logarithmically larger than the latter. (...) As a conclusion, the data-driven ATO control problem is statistically tractable, in the sense that a relatively small sample is sufficient to accurately estimate the optimal value function and a near-optimal policy.

Furthermore, we have clarified that the purpose of THEOREM 2 differs conceptually from standard optimality bounds for Fitted Q-Iteration (FQI) found in the literature (e.g., Antos et al. 2007). In Section 4.4 (pp. 17–20), we provide a detailed comparison.

(...existing analyses of FQI) generally focus on understanding the sufficient size of an NN model or the sufficient size of training data to achieve a desired accuracy. For example, with training data of N state transitions, Antos et al. (2007) shows that an FQI algorithm generates a policy of $O(N^{-1/(4p_A+4)})$ optimality gap, assuming that the optimal actor value functions can be represented by a Lipschitz class. In comparison, we take as an assumption that a properly trained NN achieves approximation accuracy ε_m and our goal is to derive an interpretable optimal gap for the specific ATO control problem in terms of the demand sample size M, usually exogenously given by ATO practitioners. Our analysis also requires a technically challenging proof that the optimal value function is Lipschitz continuous, which we discuss in Section 4.5 below.

8. **Comment:** *The proof requires a fair amount of detail in the appendix, but seems to be based on standard arguments (Contraction of Bellman operator for discounted RL problems; Coupling pairs of trajectories; and Concentration Inequalities for I.I.D. sums and covering number bounds). The paper has some amount of technical depth.*

   Response: Thank you! We have taken significant efforts to highlight our contributions in the proofs. Our goal is to prove Lipschitz continuity for the value function of an infinite-horizon problem. Unlike finite horizon problems, proving that the Bellman operator maps Lipschitz continuous functions to Lipschitz continuous functions is not sufficient.

   (a) The difficulty. In the revised manuscript, we have provided additional justification for why Lipschitz continuity is challenging to prove.

   (i) *A bibliographic remark.* We first provide a bibliographic remark on this question.

   ...we remark that the need for Lipschitz continuity is known in analyzing empirical solutions for infinite horizon MDP on continuous state spaces, see, for example, Li et al. (2021) and Gupta et al. (2024). A sufficient condition for such Lips-

chitz continuity is given by Hinderer (2005), which essentially specifies that an L-Lipschitz family with respect to a proper norm is an invariant subset of the Bellman operator.

(ii) *A counterexample.* To demonstrate that these standard arguments do not apply, we have added specific counterexamples in Section 4.5. These examples show that function classes defined by standard symmetric norms (like L^1 or L^∞) are not invariant under the Bellman operator for ATO systems.

Example 1 *Consider an ATO system, where a single product is assembled from two components with unit lead times. Assembly takes one unit of each component. There is no arriving demand. Holding and backlog costs are 1. The state and action are* $\boldsymbol{s} := ((x_1, x_2), \tilde{u}, (R_{1,1}, R_{2,1}))$ *and* $\boldsymbol{a} := (y, (z_1, z_2))$. *We consider a discount factor* $\gamma = 0.9$ *and a damping factor* $\alpha = 0.1$. *Take demand support* $\bar{d} = 1$ *so that* $(\boldsymbol{s}, \boldsymbol{a}) \in \mathcal{SA}_\alpha$ *whenever* $\|\boldsymbol{s}\|_\infty \leq 10$, $z_1, z_2 \in [0, 10]$, *and* $y \in [0, \min\{x_1, x_2, \tilde{u}\}]$.

In Example 1, Theorem 3 fails under commonly used norms and Lipschitz continuity.... Define function $w(\boldsymbol{s}, \boldsymbol{a}) := L(x_1 + x_2 + \tilde{u} + y)$, *which is* $(L, \|\cdot\|_1)$*-Lipschitz and* $(4L, \|\cdot\|_\infty)$*-Lipschitz. ...Because* $\|(\boldsymbol{s}_2, \boldsymbol{a}_2) - (\boldsymbol{s}_1, \boldsymbol{a}_1)\|_1 = 4$ *and* $\|(\boldsymbol{s}_2, \boldsymbol{a}_2) - (\boldsymbol{s}_1, \boldsymbol{a}_1)\|_\infty = 1$, *there exists no* L *such that* $\mathcal{T}^* w$ *and* w *have the same Lipschitz constant as* w. *In particular, we cannot infer, using the existing approach, that the function* Q^*_α, *which is the unique fixed point of* $\mathcal{T}^*$, *must belong to a Lipschitz class.*

(b) Necessity of ALC. In the revision, we have made additional efforts to show that the asymmetry is necessary.

(i) *An additional counterexample.* We provide an additional counterexample in Section 4.5 to indicate that a symmetric bound is not sufficient.

To further illustrate the role of asymmetry, consider seminorm

$$\rho'_1(\boldsymbol{s}, \boldsymbol{a}) := \|\boldsymbol{x} - A\boldsymbol{y}\|_1 + \|\tilde{\boldsymbol{u}} - \boldsymbol{y}\|_1 + \sum_{i=1}^{n} \|R_i\|_1 = |x_1 - y| + |x_2 - y| + |\tilde{u} - y| + |R_{1,1}| + |R_{2,1}|,$$

which is the symmetric version of ρ_1. Under seminorm ρ_1', the function w is no longer Lipschitz continuous. However, the Lipschitz constant with respect to ρ_1' again exhibits no contraction under $\mathcal{T}^*$. To show this, take $w'(\boldsymbol{s},\boldsymbol{a}) = L(x_2 - x_1 + \tilde{u} - y)\ldots$

(ii) *Intuitions.* We provide clearer justifications in Section 4.5 that a valid invariant subset must have asymmetric bounds on the slopes along different directions.

There is an intrinsic explanation that the ALC property in Theorem 3 captures the asymmetry between overage and underage in assembly systems. Specifically, in some states, a decrease in the inventory of any component can cause insufficient production and, in turn, an increase in the inventory of many other components in the next period. Therefore, among all functions in an invariant subset of the Bellman operator, the maximum slope evaluated at such states must be higher in the direction of decreased inventory compared to the direction of increased inventory, necessitating an asymmetry.

(c) Sufficiency. The revision has also provided more details on how the ALC property is used to establish the desired Lipschitz continuity and to justify its sufficiency.

It is therefore significant from Theorem 3 that, with such asymmetry captured, a proper ALC family becomes an invariant of the Bellman's operator. Using the iteration in Lemma 1, the function Q_α^* is ALC and, in turn, Lipschitz continuous. Using Lemma 1(iii), we can then prove the Lipschitz continuity of $V_\alpha^*(\,\cdot\,;\widehat{F})$. Our novel finding here is that the Lipschitz continuity, which is a symmetric notion, needs to be proved by accounting for the asymmetric behavior of the ATO system.

9. **Comment:** *The paper should define the ICNN somewhere or give a reference for rigor. There are other abbreviations that are used without introductions or definitions. It would be helpful to the readers if the authors follow the paper-writing standard that abbreviations need to be introduced before being used.*

Response: Thank you for your comment. In the revision, we have ensured that all abbreviations—including ICNN—are introduced and defined at their first appearance in the manuscript.

10. **Comment:** *The exposition of the paper is not very clear. For example, the letter M is used in various ways in different parts of the paper and in numerical experiments. The writing could be improved.*

    Response: Thank you for your suggestion. We have worked diligently to improve the paper's writing and clarity in this revision. In particular, we carefully reviewed all notation—especially symbols such as M—to ensure consistency throughout. In the revised manuscript, M is consistently used to denote the demand sample size from the introduction through the numerical section. We have also added reminders of this notation in several locations to avoid potential confusion about the size of the training data used in RL algorithms.

11. **Comment:** *Figure 1 was mentioned on Page 2 of the manuscript, but only shows up on Page 34. The tables and figures for the numerical experiments are piled up after References. It would be much preferred to include at least the important tables and figures in the main text, and surround them with explanations and analysis.*

    Response: We appreciate your concern regarding the placement of figures and tables. The revised placement follows the journal's formatting requirements, which now specify that figures must be embedded in the manuscript and tables must be included at the end of the manuscript in the submitted version. The relevant guideline is provided at the following hyperlink: Submission Guidelines (last accessed on Jan 21, 2026).

12. **Comment:** *From the current presentation of numerical experiments, it appears that the improvement is small compared with an off-the-shelf method that naively uses SAA to handle input data.*

    Response: Thank you for the comment. The improvement in optimality gap is negligible for small networks but notable for larger networks. In the revised manuscript, we have revised the paper as follows:

    (a) Clarification on the RL benchmark. We have clarified that there is no existing "*off-the-shelf method that naively uses SAA to handle input data*" for the ATO problem.

The reason is as follows.

Our design and analysis focus on achieving a computationally tractable optimization of the Q function, which is a challenging constrained minimization problem over a continuous, multidimensional, non-orthotope feasible set defined by state-dependent constraints. In particular, existing off-the-shelf RL methods are not applicable as they require the action space to be a state-independent orthotope (see, for example, Fujimoto et al. 2018).

In Section 5 (pp. 26) of the revised manuscript, we have clarified that the RL benchmark is carefully designed and tailored to the ATO problem.

Since the TD3 algorithm can not be directly applied to ATO control problems due to the state-dependent action space, we consider a minimally modified version, named after the penalized TD3 (PTD3) algorithm.

(b) Numerical comparison. In large-sized or medium-sized ATO problems, the performance gap is both statistically significant and operationally meaningful.

(i) (Large size) In the opening case study of the processed food company, we observe a critical distinction: the non-convex RL benchmark (PTD3) frequently fails to converge, regardless of hyperparameter tuning. In sharp contrast, the proposed CTD3 algorithm converges rapidly to a cost-efficient policy with minimal tuning.

 We also observe that the CTD3 algorithm can achieve satisfying policies under the learning rates from {5e-5, 1e-4, 5e-4}. In contrast, the PTD3 algorithm fails to deliver a substantial cost reduction under all choices of the learning rate.

(ii) (Medium size) In the mid-scale example, the CTD3 algorithm outperforms the RL benchmark by approximately 10% when $M = 10$, which is operationally significant. The revision clarifies that, with a tuned learning rate, the difference in optimality gap between the two algorithms at $M = 500$ and $M = 1000$ is indeed small.

> *With tuned learning rates, the difference in optimality gap between the two algorithms becomes smaller, yet still significant when $M = 10$.*

Based on these results, the revised manuscript concludes that the convexity property delivers a substantial, practically significant improvement in both final policy quality and algorithmic convergence speed.

13. **Comment:** *The paper calls Dynamic Programming "Dynamical Programming" in a few places.*

    Response: Thank you for the correction. We have corrected all instances of "Dynamical Programming" to "Dynamic Programming" in the revised manuscript.

14. **Comment:** *Page 13 should say π is a Markovian policy.*

    Response: Thank you for pointing this out. We have updated the text to state that π is a Markovian policy.

3 Response to Referee 2

Thank you! We really appreciate your kind suggestions and warm encouragement. Thank you for emphasizing the importance and challenge of solving an assemble-to-order (ATO) system. In this round of revision, we have taken your suggestions seriously and addressed them as follows.

1. **Comment:** *At a high level I think the positioning of the overall result needs to be well-specified. It seems the main technical result (i.e., the approximation bound in Theorem 2) relies on showing the ATO value function is convex (corollary 1) and that the Bellman operator preserves an asymmetric Lipschitz property (theorem 3).*

   Response: Thank you for your careful observation. We appreciate your summary of the connections between our technical results. To better specify the positioning of the overall result, we have made the following changes:

   (a) We have simplified the proof of Theorem 2 so that it now relies exclusively on the asymmetric Lipschitz continuous (ALC) property established in Theorem 3. As a result, Theorem 2 no longer depends on the convexity property from Corollary 1. This independence is now highlighted in Section 4.4.

   (b) We have reorganized the results (Lemma 1, Theorem 1, and Corollary 1) in Section 4.2. This separates the convexity properties from the analysis of the approximation bound in Theorem 2.

   We are grateful for this suggestion, as it led us to decouple these properties and strengthen the proof of our main result.

1.1 **Comment:** *As the authors state (second paragraph after Corollary 1), the convexity of the value function is not surprising (it is the classic intuition underlying optimality of base-stock policies in backlogging systems). So the contribution here appears to be using this observation together with a convex neural network to represent the high-dimensional value function.*

Response: Thank you for this insightful observation. We fully agree that convexity is the "*classic intuition underlying optimality of base-stock policies*" in demand backlogging systems. However, this property has not been fully investigated its potential to substantially accelerate the convergence of deep reinforcement learning (RL) algorithms.

In the revision, we provide a detailed explanation in Sections 4.2 and 4.6 of the convex property. Please also refer to our detailed response to the first comment by the AE (pp. 8 in this document). We summarize the main modifications below:

(a) The sharpened focus. We have sharpened the focus of the convex property by supporting *global* convergence of an optimization subroutine of the RL approach.

(b) Additional counterexamples. We show that standard Q-iteration may fail to converge in the absence of convexity in Appendix EC.3 with two additional counterexamples.

(c) Computational acceleration. We show that the convex property significantly accelerates the convergence of the RL algorithm in Sections 4.6 and 5.3.

1.2 **Comment:** *But this doesn't clearly address the high-dimensional/computational difficulty of the classic ATO problem, because, if I understand correctly, we've changed the ATO problem (which using the 15x8 example in the second paragraph of of Section 1.1 has dimension 76) to a neural network with width 256 and depth 3, i.e., still a high dimensional problem.*

Response: We agree that the use of convexity does not completely address "*the high-dimensional/computational difficulty of the classic ATO problem.*" The revision has added corresponding clarification in Section 4.2 to eliminate any overstatement.

...a Q learning algorithm approximates each w_k within a suitable function class $\mathcal{G}$. Typically, $\mathcal{G}$ a neural network (NN) class... obtaining such an approximation may be difficult in theory, especially when w_k is high-dimensional...

The benefits of employing NN are further detailed in our response to the next comment.

1.3 **Comment:** *Perhaps the well developed NN optimization procedures still have better computational performance compared to just trying to solve the dynamic program (DP) directly, even though the dimension is higher, but it is unclear that this is the case from the current paper because there is no attempt made to solve the DP.*

Response: Thank you for suggesting the benchmarking against a dynamic programming (DP) approach. To justify the benefits of the NN approach, we have added the following content:

(a) Heuristic explanation for NN approximation. In Section 4.2 of the revision, we first anchor the choice of NN in its approximation capabilities and recent success in high-dimensional control problems.

. . . (the algorithm) approximates each w_k within a suitable function class $\mathcal{G}$. Typically, $\mathcal{G}$ a neural network (NN) class as it can uniformly approximate any continuous function on a compact set (Cybenko 1989). . . . the practical success of NN on control problems of similar dimensionality (see, for example, Gijsbrechts et al. 2022 and Dai and Gluzman 2022) supports its utility to our DSAA problem. In particular, provided that $\mathcal{T}^* w_k$ can be efficiently evaluated on any state-action pair, we assume access to w_{k+1} that approximates $\mathcal{T}^* w_k$ within $\mathcal{G}$ to a desired accuracy.

Section 5 further clarifies that the NN-based RL approach can concentrate its computational resources on the most relevant regions of the state space. In contrast, standard DP methods require uniform sweeps over the entire state space.

Compared to a conventional DP solution that requires exhaustive traversal of the state space, RL algorithms focus on learning the subset of the most relevant states. It is already well recognized in reinforcement learning theory (see Chapter 9 of Sutton et al. 1998) that the targeted allocation of representational capacity in NN training not only reduces the computational burden compared to uniform DP sweeps, but also enhances accuracy in regions of the state space that most directly influence performance.

(b) DP Benchmark Implementation. Following your recommendation, we have implemented a DP benchmark with state aggregation based on Nadar et al. (2018). Please refer to our response below for the specific comparisons of cost and computation time.

1.4 **Comment:** *It also seems that the NN has the benefit of being a learning algorithm so learning directly from the data is perhaps improved, but again, it isn't clear that this should outperform a DP approach that simply uses the empirical demand distribution (i.e., an SAA DP) to solve the true ATO dynamic program, either in cost minimization or computation time.*

Response: Thank you for the suggestion. In the revised manuscript, we have implemented the suggested DP benchmark utilizing the state aggregation techniques from Nadar et al. (2018). We compared both the cost and the computational time (training/inference) of our approach against this baseline. The implementation details and results are summarized as follows:

(a) Implementation details. In Section 5.1, we introduce how the state and action spaces are discretized for computational feasibility.

We discretize the state space using a finite grid. To form the grid, k vertices are sampled on each dimension, reducing the state space $\mathcal{S}$ into a finite set of cardinality k^{ps}. We call each vertex an aggregate state. The action space is also reduced for tractability. The component replenishment decision of each component is discretized into k levels. Assembly of each product follows a priority rule. There are $m!$ different priority rules in total for m distinct products. Therefore, the reduced action space is a finite set with cardinality $k^n \times m!$. The dynamic programming (DP) algorithm is conducted by value iteration using an empirical demand distribution based on the M demand samples. Each iteration requires evaluating the state transition function over each aggregate state, action, and demand sample, counting for $k^{ps+n} \times m! \times M$ evaluations of state transitions.

(b) Comparison to the DP benchmark. In Section 5.2, we have added a comprehensive comparative analysis between our algorithm and the DP benchmark. This discussion evaluates the performance in terms of cost and computation time (training/inference time).

(i) *Cost comparison.* Our proposed RL algorithm yields a policy that significantly outperforms the DP benchmark (with 8 lattices). While increasing the lattice resolution could theoretically improve the DP benchmark's performance, the corresponding training time grows prohibitively (as detailed in point (ii)).

In the N-system with independent demand (using 200 samples), the DP benchmark incurs an optimality gap of 9.3% with eight lattices, whereas our algorithm achieves a gap of only 1.4%. Similarly, under correlated demand, our algorithm achieves a significantly lower cost (58.1) compared to the DP benchmark (67.9). For ease of reference, the full results are presented in Table 2 below.

	Algorithm	Demand Sample Size		
		$M = 10$	$M = 100$	$M = 200$
		C (s.d.)	C (s.d.)	C (s.d.)
I.I.D. Demand (Optimality: 81.0)	CTD3	86.9 (0.04)	82.9 (0.06)	82.2 (0.04)
	PTD3	87.4 (0.04)	84.6 (0.05)	83.0 (0.04)
	NV-PRP	86.1 (0.05)	84.7 (0.03)	82.9 (0.04)
	DP (4 Lattices)	105.6 (0.08)	105.5 (0.09)	105.3 (0.06)
	DP (6 Lattices)	100.8 (0.07)	100.9 (0.07)	97.4 (0.06)
	DP (8 Lattices)	99.7 (0.04)	88.9 (0.06)	88.5 (0.05)
Correlated Demand	CTD3	66.1 (0.03)	60.2 (0.05)	58.1 (0.02)
	PTD3	67.3 (0.03)	62.6 (0.04)	61.1 (0.05)
	NV-PRP	75.8 (0.06)	78.9 (0.07)	73.7 (0.07)
	DP (4 Lattices)	92.1 (0.04)	92.1 (0.06)	92.0 (0.04)
	DP (6 Lattices)	75.0 (0.07)	75.0 (0.03)	74.9 (0.05)
	DP (8 Lattices)	70.4 (0.05)	70.4 (0.04)	67.9 (0.03)

Table 2: The performances of CTD3 policy, PTD3 policy, NV-PRP policy, and DP policies on the N-system with different demand sample sizes and demand joint distributions.

(ii) *Computation time comparison.* As shown in Table EC.3, the training time for the DP benchmark grows exponentially with the lattice resolution in the N-system. For a sample size of $M = 200$, the 4-lattice DP is fast, but increasing the resolution to 8 lattices raises the training time to 221,687 seconds. In sharp contrast, our CTD3 algorithm requires only 4,420 seconds, which is approximately 50 times faster than the high-resolution DP benchmark. Furthermore, the inference time for our algorithm (0.49 ms) remains comparable to that of the DP benchmark (0.43 ms), ensuring real-time applicability.

We also note that the DP benchmark is computationally infeasible for the two larger ATO examples.

It is worth noting that (for the ATO system in Example 3) the computational com-

	Algorithm	Demand Sample Size		
		$M = 10$	$M = 100$	$M = 200$
Training Time (s)	CTD3	2482	4035	4420
	PTD3	4980	6311	9856
	DP (4 Lattices)	53	385	738
	DP (6 Lattices)	1434	10937	19137
	DP (8 Lattices)	14012	107580	221687
Inference Time (ms)	CTD3		0.49	
	PTD3		0.48	
	NV-PRP		2.09	
	DP (4 Lattices)		0.43	
	DP (6 Lattices)		0.42	
	DP (8 Lattices)		0.43	

Table EC.3: Training time and inference time comparison for N-system. Inference time is averaged over 1,000 steps.

plexity of the DP-based state aggregation method becomes astronomical even with only four vertices per dimension, with nearly 10^{16} *evaluations of state transition required for each value iteration, given* $M = 10$ *demand samples.*

(iii) *Intuitions.* In the revision, we also provide analytical evidence on why RL is preferred over solving DP directly. The exhaustive traverse of the state space results in huge computational complexity of the DP benchmark.

With eight vertices in each dimension, there are 262,144 aggregated states. The reduced action space contains 128 actions. With 200 demand samples, each DP iteration requires evaluating approximately 10^{10} *state transitions, and the computational time easily exceeds that of the RL algorithms. Because our testing environment cannot accommodate further increases in the number of vertices per dimension due to computational capacity constraints, we conclude that the RL algorithms significantly outperform the state aggregation algorithm under reasonable computational power.*

The computational intractability of the DP benchmark necessitates a coarse state aggregation (i.e., fewer lattices). This restriction introduces significant discretization error, which directly translates into a substantial performance gap compared to the RL algorithm.

1.5 **Comment:** *Also, the assymetric Lipschitz property is interesting, but it not entirely clear why it is needed from the discussion in the paper.*

Response: Thank you for your suggestion to expand the discussion of the ALC property, which has significantly strengthened the paper. In the revision, we have included multiple counterexamples and additional remarks on the literature to demonstrate the necessity of the ALC property in our analysis.

To justify "*why (the ALC) is needed from the discussion in the paper,*" the revision has added bibliographical remarks and counterexamples to demonstrate the inapplicability of existing approaches.

(a) Bibliographical context. The ALC argument is used to establish the Lipschitz continuity of the optimal value function. In the revised manuscript, we have added an extensive bibliographic remark noting that the need for Lipschitz continuity is well known, while the proof can be challenging.

...we remark that the need for Lipschitz continuity is known in analyzing empirical solutions for infinite horizon MDP on continuous state spaces, see, for example, Li et al. (2021) and Gupta et al. (2024). A sufficient condition for such Lipschitz continuity is given by Hinderer (2005), which essentially specifies that an L-Lipschitz family with respect to a proper norm is an invariant subset of the Bellman operator.

(b) Counterexamples.

To substantiate our ALC approach, the revised manuscript provides counterexamples in Section 4.5 to demonstrate that the condition in Hinderer (2005), i.e., the existence of a Lipschitz class invariant under the Bellman operator, fails for ATO systems.

Example 1 *Consider an ATO system, where a single product is assembled from two components with unit lead times. Assembly takes one unit of each component. There is no arriving demand. Holding and backlog costs are 1. The state and action are* $\boldsymbol{s} := ((x_1, x_2), \tilde{u}, (R_{1,1}, R_{2,1}))$ *and* $\boldsymbol{a} := (y, (z_1, z_2))$. *We consider a discount factor* $\gamma = 0.9$ *and a damping factor* $\alpha = 0.1$. *Take demand support* $\bar{d} = 1$ *so that* $(\boldsymbol{s}, \boldsymbol{a}) \in \mathcal{SA}_\alpha$ *whenever* $\|\boldsymbol{s}\|_\infty \leq 10$, $z_1, z_2 \in [0, 10]$, *and* $y \in [0, \min\{x_1, x_2, \tilde{u}\}]$.

In Example 1, Theorem 3 fails under commonly used norms and Lipschitz continuity.... Define function $w(\boldsymbol{s}, \boldsymbol{a}) := L(x_1 + x_2 + \tilde{u} + y)$, which is $(L, \|\cdot\|_1)$-Lipschitz and $(4L, \|\cdot\|_\infty)$-Lipschitz. ... Because $\|(\boldsymbol{s}_2, \boldsymbol{a}_2) - (\boldsymbol{s}_1, \boldsymbol{a}_1)\|_1 = 4$ and $\|(\boldsymbol{s}_2, \boldsymbol{a}_2) - (\boldsymbol{s}_1, \boldsymbol{a}_1)\|_\infty = 1$, there exists no L such that $\mathcal{T}^* w$ and w have the same Lipschitz constant as w. In particular, we cannot infer, using the existing approach, that the function Q^*_α, which is the unique fixed point of $\mathcal{T}^*$, must belong to a Lipschitz class.

(c) Intuitive explanation. We also provide further intuitive explanation of the ALC property in this revision.

(i) An additional counterexample indicates that a symmetric bound on the slopes is not sufficient, necessitating the asymmetry.

To further illustrate the role of asymmetry, consider seminorm

$$\rho'_1(\boldsymbol{s}, \boldsymbol{a}) := \|\boldsymbol{x} - A\boldsymbol{y}\|_1 + \|\tilde{\boldsymbol{u}} - \boldsymbol{y}\|_1 + \sum_{i=1}^{n} \|R_i\|_1 = |x_1 - y| + |x_2 - y| + |\tilde{u} - y| + |R_{1,1}| + |R_{2,1}|,$$

which is the symmetric version of ρ_1. However, the Lipschitz constant with respect to ρ'_1 again exhibits no contraction under $\mathcal{T}^*$. To show this, take $w'(\boldsymbol{s}, \boldsymbol{a}) = L(x_2 - x_1 + \tilde{u} - y)$... Again, there is no L such that $\mathcal{T}^* w'$ is (L, ρ'_1)-Lipchitz.

(ii) The revision now provides a clearer justification that a valid invariant subset must have asymmetric bounds on the slopes along different directions.

There is an intrinsic explanation that the ALC property in Theorem 3 captures the asymmetry between overage and underage in assembly systems. Specifically, in some states, a decrease in the inventory of any component can cause insufficient production and, in turn, an increase in the inventory of many other components in the next period. Therefore, among all functions in an invariant subset of the Bellman operator, the maximum slope evaluated at such states must be higher in the direction of decreased inventory compared to the direction of increased inventory, necessitating an asymmetry.

(iii) With the same example, we show that the asymmetric seminorm captures the asymmetry.

In the example, an overage of component 1 costs one unit from holding the component in the inventory, an underage of component 1 costs two units, one from holding component 2 in the inventory and the other from backlogged demand due to insufficient production. Under asymmetric seminorm ρ_1, an L-ALC function can correspondingly increase by $2L$ when x_1 decreases by 1, but can increase by at most L when x_1 increases by 1.

1.6 **Comment:** *The authors point out that the slope of the value function is asymmetric when moving the inventory in different directions, but this is not a novel issue in inventory management, and it isn't clear why we can't just consider the maximum of the two slopes?*

Response: We agree that it is not conceptually novel that the (optimal) value function has different slopes "*when moving the inventory in different directions.*" The novelty here is that the Lipschitz continuity of the value function, which requires one bound on the slopes in both directions, must be proved by employing different bounds in different directions. Once both slopes are bounded, we indeed prove that the optimal value function is Lipschitz continuous by taking "*the maximum of the two slopes.*" The revision now clarifies this fact in Section 4.2, providing further justification for the novelty of the proof.

It is therefore significant from Theorem 3 that, with such asymmetry captured, a proper ALC family becomes an invariant of the Bellman's operator. Using the iteration in Lemma 1, the function Q_α^* is ALC and, in turn, Lipschitz continuous. Using Lemma 1(iii), we can then prove the Lipschitz continuity of $V_\alpha^*(\,\cdot\,;\widehat{F})$. Our novel finding here is that the Lipschitz continuity, which is a symmetric notion, needs to be proved by accounting for the asymmetric behavior of the ATO system.

1.7 **Comment:** *So, the high level theoretical idea of Theorem 1 seems to be that a convex NN (or other regressor function) can be used to learn/approximate the optimal value function of ATO problems, but it is not entirely clear that this is an improvement over classical DP, or entirely novel, because the convexity is standard and the NN may also be high-dimensional.*

Response: Thank you for your sharp observations, which motivated us to improve the exposition and enhance our numerical benchmarks.

First, regarding the improvement "*over classical DP*," we provide a thorough benchmarking in our response to your **Comment 1.4**, demonstrating that our proposed algorithm overcomes the computational intractability that limits standard DP methods.

Second, regarding the "*high level theoretical idea of Theorem 1 (convexity)*," we provide a focused discussion on the benefits of convexity in Sections 4.2 and 4.6. As summarized in our response to your **Comment 1.1**, we highlight that integrating the convex structure is essential for our RL algorithmic design that provably converges to a near-optimal policy.

2. **Comment:** *An obvious way to address the above concern would be through the numerical simulations, by comparing computation time and performance to the DP solution or to heuristics from the literature. Unfortunately I think the paper falls short in this regard on multiple dimensions.*

   Response: Thank you for the comments. In the revision, we include direct benchmarking against a DP approach with state aggregation. As detailed in the responses below, we found that these additional analyses substantially strengthen the justification for our algorithm.

2.1 **Comment:** *First, I cannot decipher a clear comparison of computation times from the results reported. Training NNs is typically computationally intensive, so it would be helpful to see the total training time for the NN vs. the time it takes to compute the simpler heuristics, such as the newsvendor (NV) heuristic, which one would expect to be quite fast. If the NN takes significantly longer, one could still make a case in many practical settings for using the simpler heuristic if speed is more important.*

Response: Thank you for raising this important practical consideration. We agree that "*training NNs is typically computationally intensive*". In the revision, we provide a detailed comparison of computation times in Appendix EC.5.

In our response to **Comment 1.4**, we presented the quantitative comparison in Table EC.3. A summary of these results is as follows:

- *Comparison to nonconvex RL:* The proposed CTD3 algorithm converges in noticeably shorter training time than the nonconvex PTD3 benchmark, confirming the algorithmic benefits of the convexity-based design.
- *Comparison to DP benchmark:* The overall training time of the CTD3 algorithm is shorter than that of the DP benchmark with the finest lattice resolution.
- *Comparison to NV-PRP benchmark:* While the NV-PRP heuristic requires zero training time, its inference time is actually higher than that of the RL policy (2.09 ms vs 0.49 ms). This is because the heuristic must solve an optimization problem at every decision epoch, whereas the trained RL policy simply executes a fast forward pass (often GPU-accelerated).

A more detailed discussion from Appendix EC.5. is provided below.

We observe that the DP benchmark is computationally efficient at coarse discretization levels; however, its training time escalates rapidly as the resolution (number of lattice points) increases, eventually exceeding that of the RL algorithms. Furthermore, the CTD3 algorithm converges significantly faster than PTD3, empirically validating the role of convexity in accelerating convergence. Note that we do not report training time for the NV-PRP heuristic, as it does not require training.

Regarding inference time, both the DP and RL algorithms exhibit comparable performance and significantly outperform the heuristic baseline. This is because RL inference can be accelerated on a GPU, while DP actions can be retrieved from a hash table with $O(1)$ *complexity. In contrast, the NV-PRP heuristic must compute order quantities and assembly priorities at every decision epoch, resulting in higher latency.*

We believe that the training time does not preclude the practical use of RL algorithms. Below we provide a non-exhaustive list:

(i) *Utility in retrospective analysis.* RL policies can serve as a valuable benchmark for practitioners. Even when not deployed directly for execution, they provide a baseline to evaluate the performance of existing heuristic policies and to quantify potential gaps.

(ii) *Practical acceleration in industrial settings.* Industrial RL implementations can be engineered for substantially faster convergence. For instance, practitioners can restrict the policy class to a lower-dimensional family tailored to the specific application, which may reduce training complexity and computational time.

(iii) *Fast inference after training.* While training RL policies is computationally intensive, evaluating a trained policy is typically negligible. In our processed-food industry example, training requires approximately 16 hours, whereas computing the action for a single state takes only 0.6 ms. Consequently, firms can rely on fast, repeated inference for daily operations while limiting retraining to infrequent intervals. Furthermore, transfer-learning techniques can be employed to significantly accelerate convergence when fine-tuning an existing model.

2.2 **Comment:** *Another lacking comparison is with the DP solution. The problem is a well-posed DP if we assume the empirical demand distribution, and the state space for some of the systems considered isn't too large to solve directly, so a comparison with this approach could be made. Perhaps the 15x8 system becomes too large when considering all states in 76 dimensions, but it seems the N-system at least could be solved in some settings, and regardless, a discussion of this issue would also help motivate why the NN approach is actually needed.*

Response: Thank you for the suggestion to model the problem as a DP using the empirical demand distribution. Following this recommendation, we implemented a DP benchmark with state aggregation to facilitate a direct comparison.

As discussed in our response to **Comment 1.4**, this comparison demonstrates that the CTD3 algorithm outperforms the DP benchmark with the finest lattice resolution in terms of both solution quality and training time.

2.3 **Comment:** *Further, even if the DP approach can't be implemented, there are other* *heuristics from the literature that should be tried as benchmarks. In particular, there are stochastic programming based policies of Reiman and Wang (2015), Reiman et al. (2023), and DeValve and Myles (2025) that could solve SAA stochastic programs just using the samples you consider, as well as state space aggregation (Nadar et al. (2018)). Again, time and performance characterizations should be made against these types of policies to justify that the computation time and improved performance are worth it.*

Response: Thank you for suggesting more computational methods for solving an ATO system. We have implemented the state aggregation method (Nadar et al. (2018)), which we believe as the most suitable method.

We have investigated the work by Reiman and Wang (2015) and DeValve and Myles (2025). Unfortunately, **these works assume identical lead time for all components**. For this reason, they cannot be directly applied to our setting.

Reiman et al. (2023) generalize the policy in Reiman and Wang (2015) from homogeneous component lead times to heterogeneous lead times. Nevertheless, we find that this policy assumes the no-holdback rule, which is shown to be suboptimal in our numerical examples (see Figure 3 in the manuscript). More importantly, this policy still requires solving a multi-dimensional DP.

We appreciate this learning opportunity. We have emphasized the importance of further adapting these methods as an important future research direction.

Finally, integrating insights from stochastic programming based policies (e.g., DeValve and Myles (2025)) into RL algorithm design may also be a potential avenue for future research.

3. **Comment:** *I also have a comment on the decay approach to approximating the value function. It's a nice idea and clever use of the decay to make the state/action space compact to apply known results (assuming I am correct in interpreting Theorem 2 that it is approximating the true value function of the non-decayed problem, and that the dependence on the decay parameter shows up as $\sqrt{\log(1/\alpha)}$ which gives a nice understanding of how this approach works).*

Response: Thank you for your encouraging comment. The damping factor (previously referred to as the decay factor) functions exactly as described in your comment. We appreciate your acknowledgment that introducing the damping factor is a valid and appropriate step in our proof.

3.1 **Comment:** *However, I also want to point out that it's typically the case in inventory problems that bounding the state/action space of an optimal policy is possible directly; the holding cost and backlog costs make it so an optimal policy never lets the inventory or backlog get too high, and this can often be shown with a newsvendor type argument. See e.g., Reiman and Wang (2015) for a result bounding inventory positions in an ATO system and Huh et al. (2009) for a result in a lost sales system. It seems a similar type of argument should be possible here to remove the reliance on the decay approach.*

Response: Thank you for suggesting these alternative approaches and for directing us to the relevant results in Reiman and Wang (2015) and Huh et al. (2009).

Following your recommendation, we carefully examined whether similar arguments could be adapted to our setting. While these techniques are highly effective in many inventory and ATO models, we were not able to identify a direct adaptation that preserves the validity of our subsequent results and proofs in the present formulation. We provide the detailed reasoning below.

(a) Preservation of structural properties.

To apply the necessary concentration inequalities in our main proof, we require three simultaneous conditions: (i) a **bounded** state space, (ii) **linearity** in the transition dynamics, and (iii) **convexity** of the state-action space. As detailed in the expanded Proposition 1 in Section 4.1 (pp. 13–14), the damping approach satisfies all three requirements:

Proposition 1[Boundedness] *The following results hold.*

(i) For any $\alpha \in (0,1)$, f_α *is a linear function and* $\mathcal{SA}_\alpha$ *is a bounded convex set.*

(ii) For any $\alpha \in (0,1)$, $f_\alpha(\boldsymbol{s}, \boldsymbol{a}, \boldsymbol{d}) \in \mathcal{S}_\alpha$ *holds whenever* $(\boldsymbol{s}, \boldsymbol{a}) \in \mathcal{SA}_\alpha$ *and* $\boldsymbol{d} \in \mathcal{D}$.

(iii) As $\alpha \to 0^+$, x_α, u_α, R_α, y_α, *and* z_α *increases without bound.*

Proposition 1 suggests that the damped transition function f_α preserves the linearity of f and the damped state-action space $\mathcal{SA}_\alpha$ preserves the convexity of $\mathcal{SA}$. These two properties are crucial in subsequent results.

(b) Incompatibility of standard alternatives. We explored standard alternative methods to remove the damping factor. However, we found that these methods inherently violate the structural properties established in Proposition 1:

(i) *Truncation.* Imposing a hard boundary on the state space (e.g., $\min\{f(\cdot), K\}$) ensures boundedness but destroys the **linearity** of the transition function f.

For example, truncating the state transition by $\boldsymbol{s}(t+1) = \min\{f(\boldsymbol{s}(t), \boldsymbol{a}(t), \boldsymbol{d}(t)), K\}$ for some $K > 0$ yields a bounded state space but destroys the linearity of f.

(ii) *Action restriction.* Restricting actions to "reasonable" subsets (as in Reiman and Wang (2015)) provides a more promising alternative towards an upper bound of the trajectory. Theorem 2 in the reference shows that any optimal solution to a stochastic programming lower bound remains bounded by a constant M. Lemma 5 in the reference proves that the same constant bounds the proposed asymptotically optimal solution.

Conceptually, the transition trajectory of the original problem can be bounded if one can find a bounded convex subset $\Omega \subseteq \mathcal{SA}_\alpha$ satisfying the following two properties: (i) the optimal control to the ATO system, at any state $\boldsymbol{s}$, takes optimal action $\boldsymbol{a}$ such that $(\boldsymbol{s}, \boldsymbol{a}) \in \Omega$; (ii) for any tuple $(\boldsymbol{s}, \boldsymbol{a}) \in \Omega$ and $\boldsymbol{d} \in D$, $s' = f(\boldsymbol{s}, \boldsymbol{a}, \boldsymbol{d})$ lies in the projection of Ω to $\mathcal{S}$.

Despite extensive exploration, we were unable to identify such a set Ω. In particular, the **convexity** of Ω enforces a strict condition that is not easy to satisfy. For example, similar choices as Reiman and Wang (2015) violate the convexity of Ω. In the revision, we add a brief discussion in Section 4.1.

Restricting actions to "reasonable" ones, as in Reiman and Wang (2015), can bound the state trajectory under the optimal policy but may lead to a non-convex *state-action space.*

(iii) *Coupling.* Huh et al. (2009) provides a coupling argument, making it possible to bound the trajectory of a lost-sale system by that of a backorder system. However, after a few attempts, we believe that such coupling, if it exists, would be highly nontrivial and likely more complex than our current damping approach.

(iv) *Summary.* Consequently, we retain the damping factor not merely for analytical convenience, but because it represents the minimal structural modification that ensures boundedness while preserving the linearity and convexity required for our theoretical analysis.

4. **Comment:** *Finally, the writing of the paper should be improved if it were to move toward publication. I'll refrain currently from giving a complete list of issues, but a few concerns include*

   Response: Thank you for your valuable suggestion. We have made a concerted effort to improve the presentation and writing quality throughout the manuscript. In the revision, we have expanded the discussion to better explain our technical contributions and the rationale behind each step of the proposed algorithm. Additionally, to streamline the flow, we have removed material that was not central to the primary narrative. We believe the paper is now much clearer and more focused.

4.1 **Comment:** *Page 14 line 53, why does the function class necessitate Q iteration with NN? And further in that paragraph, why is the family $\mathcal{C}$ "small"?*

Response: Upon reflection, we determined that this paragraph was tangential to the central narrative of the paper. To ensure a clearer and more focused exposition, we have removed this discussion in the revision.

4.2 **Comment (cont'd):** *On page 15, line 14, why would we expect that the ATO value function class should have to include discontinuous elements when the value function is convex?*

Response: Consistent with our response to the previous comment, we determined that this paragraph was also tangential to the core analysis. We removed it to maintain a focused, coherent narrative.

4.3 **Comment:** *On page 15 line 53, how can the function Λ be bounded from above and approach infinity?*

Response: We have clarified in the revision that Λ is defined on the extended domain $\mathcal{S}_\alpha \times \mathbb{R}^{p_A}$. Its limiting behavior depends on whether the point lies within the feasible set as $\eta \to 0^+$:

(i) If $(\boldsymbol{s}, \boldsymbol{a}) \in \mathcal{SA}_\alpha$, then $\Lambda(\boldsymbol{s}, \boldsymbol{a}; \eta) \to 0$.

(ii) If $(\boldsymbol{s}, \boldsymbol{a}) \notin \mathcal{SA}_\alpha$, then $\Lambda(\boldsymbol{s}, \boldsymbol{a}; \eta) \to +\infty$.

As $\eta \to 0^+$, the penalty Λ impose diminishing penalty inside $\mathcal{SA}_\alpha$ but increasingly large penalty outside $\mathcal{SA}_\alpha$. As a result, for sufficiently small η, the minimizer of (15) is an interior point of $\mathcal{SA}_\alpha$. Therefore, a practical algorithmic implementation can drop the constraint $\boldsymbol{a} \in \mathcal{A}_\alpha(\boldsymbol{s})$, reducing (15) to an unconstrained optimization problem on $\mathbb{R}^{p_A}$, which can be efficiently solved using gradient-type algorithms. For more stable convergence, (16) takes a diminishing control parameter η^k in iteration k.

In Section 4.3, we have added references to the interior point method literature to provide context for this barrier function design.

Algorithm 1 adopts an interior point method (IPM), a well-known convex optimization tool, to simplify the evaluation of $\mathcal{T}^*$, see, for example, Boyd and Vandenberghe (2004), Chapter 11.

4.4 **Comment:** *As well as grammatical issues/typos. I suggest the paper be proofread carefully to ensure all points made are clear and concise.*

Response: Thank you for your comment. We have made significant efforts to proofread the manuscript carefully. Grammatical issues and typos have been corrected, and the text has been revised to ensure clarity and conciseness throughout.

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